Toronto Metropolitan University

Metropolitan Undergraduate Engineering Society

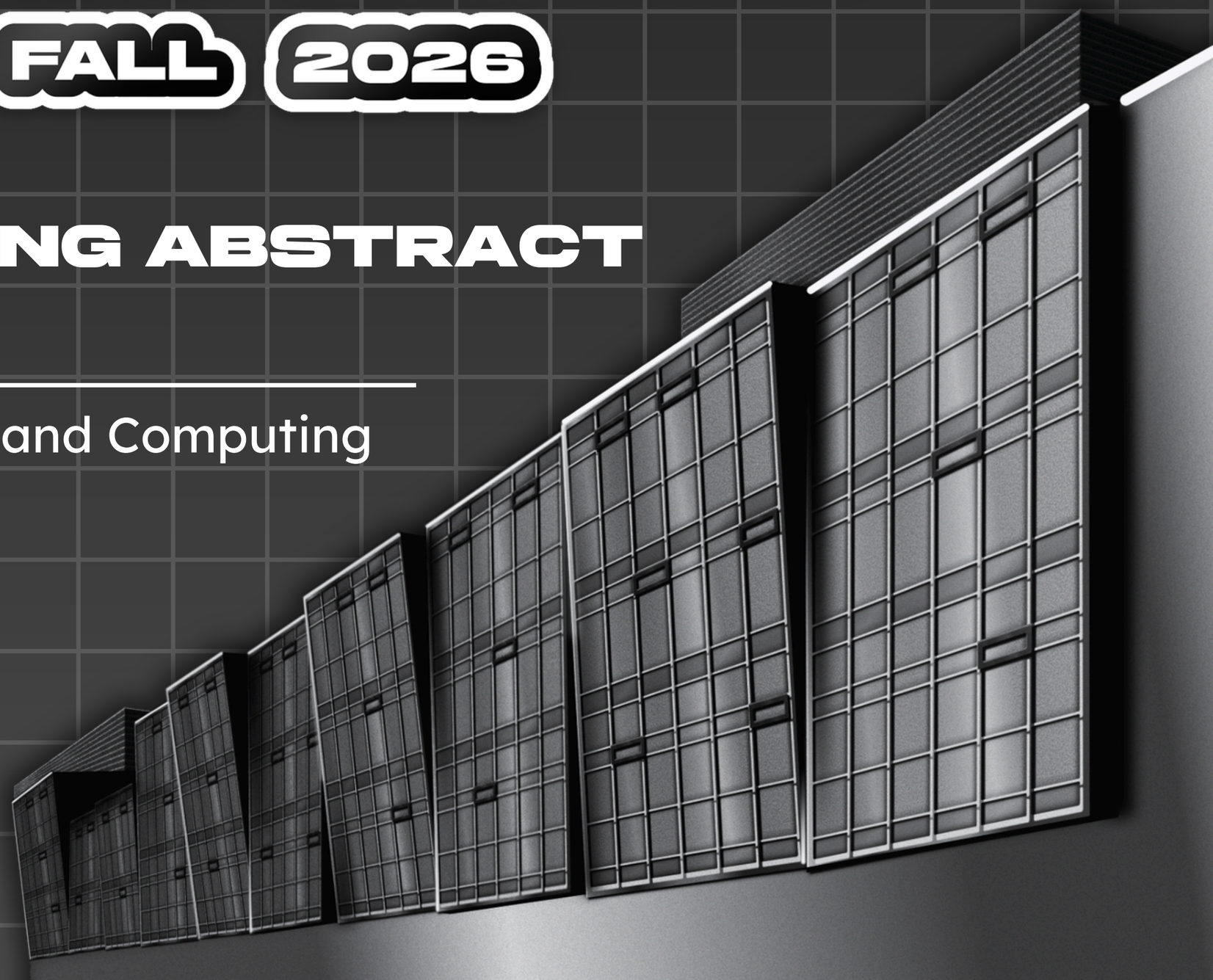
TORONTO METROPOLITAN UNIVERSITY ENGINEERING COMPETITION

FALL 2026

RE-ENGINEERING ABSTRACT

NOVEMBER 7 & 8, 2026

George Vari Engineering and Computing Centre



Metropolitan Engineering Competition 2026
Re-Engineering
Competition Package

1. INTRODUCTION

Welcome to the Metropolitan Engineering Competition (MEC 2026)! The Re-Engineering competition challenges participants to apply their engineering knowledge, creativity, and problem-solving skills to improve existing systems and address real-world challenges.

This year's theme, “Feeding the Future: Designing Smart Infrastructure for Urban Food Security,” focuses on developing practical and innovative engineering solutions that strengthen food accessibility, sustainability, and infrastructure.

In this competition, participants will analyze existing food recovery and distribution systems, identify inefficiencies and limitations, and propose engineering modifications to improve their overall performance. Teams are expected to demonstrate how their redesigns enhance efficiency, reliability, sustainability, and accessibility while considering realistic implementation constraints.

The competition consists of two cases. Case 1 will be released six days before the competition, allowing teams to conduct research and begin developing their proposed redesign. Case 2 will be revealed during the competition briefing on Saturday, November 7, 2026.

Teams will have eight (8) hours on competition day to complete their work, finalize their solutions, and prepare their submissions.

Competition Information

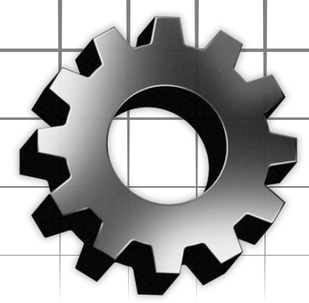
Team Size: Two (2) competitors

Competition Day: Saturday, November 7, 2026

Working Time: Eight (8) hours

Presentation Day: Sunday, November 8, 2026

Case 1 will be released six days before the competition. Case 2 will be revealed during the competition briefing on November 7. Both cases must be completed and submitted together by the official deadline announced by the Competition Coordinators.



2. DELIVERABLES

Submission Format: Teams must submit one combined final report in PDF format and one presentation file covering both cases. Presentation slides may be submitted in PDF or PowerPoint format, subject to the official submission instructions. Both files must be submitted electronically by the announced deadline.

2.1 Report

The report should clearly explain the existing systems, identify their weaknesses, and present the proposed engineering modifications and expected improvements.

The report must include:

- **Existing System Analysis:** An explanation of the current systems, including major weaknesses, inefficiencies, and limitations.
- **Proposed Redesign:** A detailed description of the engineering modifications proposed for each case.
- **System Diagrams:** Original and redesigned system or process diagrams illustrating the proposed changes.
- **Performance Comparison:** A before-and-after comparison using relevant performance indicators, calculations, or justified estimates.
- **Implementation Considerations:** Major assumptions, constraints, costs, risks, and potential implementation challenges.
- **Engineering Justification:** An explanation of how the proposed modifications improve efficiency, sustainability, accessibility, and reliability.
- Each team must submit one final report covering both Case 1 and Case 2.
- The final report must not exceed two (2) pages of main content, excluding the title page, references, and appendices. The two-page limit applies to the combined report covering both cases, not to each case individually. Appendices may contain supporting diagrams, calculations, or technical evidence but must not be used to bypass the main report's page limit.

Teams are encouraged to support their redesigns with appropriate engineering calculations, diagrams, and referenced information.

Both Case 1 and Case 2 must be included in the final report and submitted together on Saturday, November 7, 2026.

The exact submission deadline and method will be communicated by the Competition Coordinators.

2.2 Presentation

Each team must prepare a final presentation summarizing its proposed redesigns for both Case 1 and Case 2.

The presentation should clearly communicate the engineering problem, proposed modifications, expected improvements, and implementation considerations.

Teams should explain:

- The major weaknesses identified in the existing systems.
- The engineering modifications proposed for each case.
- How the redesigns improve system performance.
- Relevant performance comparisons and supporting calculations.
- Major risks, trade-offs, and implementation considerations.
- How the proposed solutions support sustainable and accessible food systems.

Both registered team members are expected to participate meaningfully in the presentation and demonstrate an understanding of the proposed redesigns. Teams should allocate sufficient presentation time to explain both cases, including their engineering decisions, expected improvements, and implementation considerations.

Presentation Duration: 15 minutes

Question-and-Answer Period: 7 minutes

3. RULES AND GUIDELINES

3.1 Allowed Resources

Competitors may use appropriate engineering software, research materials, and reference sources to support their analysis and proposed redesigns.

All external information, data, diagrams, and referenced materials must be appropriately cited.

Teams are expected to develop their own engineering analysis and must be able to explain and justify their proposed solutions during judging.

All competitors must follow the official MEC 2026 rules regarding permitted resources, generative AI, and collaboration.

3.2 Question Period

A question period will be provided during the competition briefing to allow competitors to ask questions regarding the problem statements, deliverables, rules, and judging requirements.

Competitors may approach the Competition Coordinators for clarification during the competition.

Any clarifications that affect competition requirements will be communicated consistently to all teams.

3.3 Presentation Order

The presentation order will be determined by the Competition Coordinators and communicated to all teams.

Teams must submit their final report and presentation by the announced deadline and be prepared to present at their assigned time.

Once the final submission deadline has passed, teams may not modify their submitted report or presentation materials unless explicitly authorized by the Competition Coordinators. Presentations must be based on the submitted work, although teams may provide verbal clarification and answer judges' questions during the presentation and Q&A period.

All team members are expected to be present during their assigned presentation and judging period.

4. CASE 1 PROBLEM STATEMENT

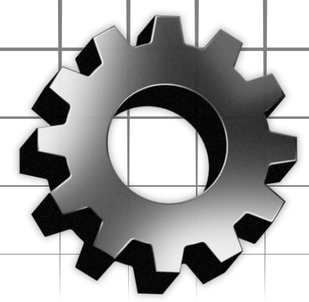
Smart Food Recovery & Waste Reduction Network

Across Toronto, edible surplus food may go unused or be discarded while many communities continue to face challenges accessing affordable and nutritious food. Existing food recovery networks rely on coordination between food producers, retailers, restaurants, food banks, community organizations, and distribution hubs. However, inefficient tracking, transportation delays, storage limitations, and poor coordination can reduce the effectiveness of these systems.

Your task is to re-engineer an existing food recovery and redistribution system in Toronto to improve how safe, edible surplus food is identified, collected, transported, stored, and distributed to communities in need.

Objectives:

- Reduce edible food waste and unnecessary landfill disposal.
- Improve the identification and tracking of safe surplus food.
- Strengthen coordination between food suppliers, food banks, community organizations, and distribution hubs.
- Improve recovery speed and transportation efficiency.
- Increase access to affordable and nutritious food.
- Reduce transportation-related environmental impacts where practical.
- Ensure the redesigned system remains scalable and compatible with existing urban infrastructure.



Teams should balance innovation, sustainability, cost, and practicality while proposing realistic improvements to existing food recovery and redistribution processes.

Note: Case 1 is released six days before the competition. Teams may use this preparation period to research and develop their redesign. No advance submission is required.

5. CASE 2 PROBLEM STATEMENT

Emergency Food Distribution Redesign

During emergencies, disruptions to transportation networks, storage facilities, and food distribution infrastructure can significantly affect access to essential food supplies. Power outages, damaged infrastructure, unavailable delivery routes, and increased demand can create additional challenges for organizations responsible for recovering, storing, and distributing food.

These disruptions may lead to food spoilage, delivery delays, inefficient resource allocation, and reduced access to food for vulnerable communities and emergency shelters.

Your task is to **re-engineer an existing emergency food distribution system** to maintain critical food access when normal infrastructure and operations are disrupted.

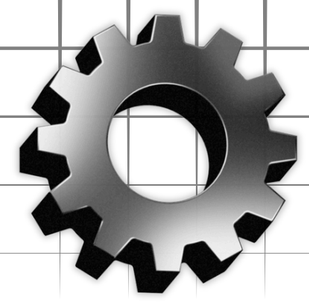
Using the scenario and constraints provided during the competition briefing, teams must identify weaknesses in the existing emergency distribution process and propose practical engineering modifications that improve reliability, resilience, and resource utilization.

Objectives:

- Maintain access to essential food supplies during infrastructure disruptions.
- Protect recovered and perishable food during power outages or storage failures.
- Develop alternative transportation and distribution methods when normal routes or facilities are unavailable.
- Prioritize vulnerable communities and emergency shelters.
- Manage increased demand and limited food inventory.
- Establish backup procedures for critical system failures.
- Develop a practical strategy for restoring normal food distribution operations.
- Ensure the redesigned system remains cost-conscious, scalable, and feasible.

Emergency Scenario and Operational Constraints

A severe storm has caused widespread power outages and transportation disruptions across Toronto. Several food storage and distribution facilities have lost electricity, while flooding and road closures have restricted access to certain delivery routes.



As a result, the existing food distribution network is unable to operate at its normal capacity. Food recovery organizations must prevent the spoilage of perishable food, maintain essential deliveries, and respond to increased demand from vulnerable communities.

Your team has been assigned to redesign the emergency food distribution process using the following standardized information.

1. Existing Distribution Network

The affected network consists of:

- One (1) central food distribution hub.
- Three (3) community distribution centres.
- Four (4) delivery vehicles.
- A combination of refrigerated and non-refrigerated food storage.

The central hub remains operational but has limited backup electricity. One community distribution centre is temporarily inaccessible because of flooding.

2. Available Food Inventory

The network currently has:

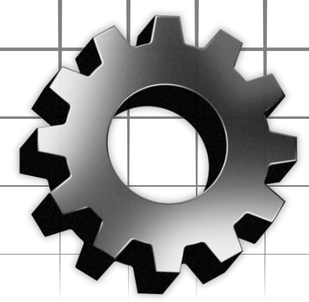
- 2,000 kg of perishable food requiring temperature-controlled storage.
- 3,000 kg of non-perishable food.
- Refrigerated storage capacity for only 1,000 kg of perishable food during the outage.

Teams must determine how to protect, prioritize, redistribute, or otherwise safely manage the remaining 1,000 kg of perishable inventory that exceeds the available refrigerated storage capacity. Proposed solutions must consider appropriate food safety requirements, available transportation resources, and the feasibility of implementing emergency storage or redistribution measures within the scenario constraints.

3. Transportation and Distribution Constraints

Of the four delivery vehicles:

- Two (2) vehicles remain fully operational.
- One (1) vehicle is unavailable due to mechanical failure.
- One (1) vehicle remains mechanically operational but cannot access its usual delivery route because of flooding. Teams may propose an alternative route or reassignment for this vehicle, provided the proposed solution is practical under the emergency conditions.



Normal delivery routes have experienced significant delays. Teams may propose alternative routes, vehicle-sharing arrangements, temporary distribution locations, or other practical modifications.

4. Emergency Food Demand

The network must provide emergency food assistance to approximately 1,500 people over the next 24 hours.

Priority must be given to:

- Emergency shelters.
- Vulnerable individuals and communities.
- Areas experiencing limited access to food distribution services.

Teams must explain how available food and transportation resources will be allocated fairly and efficiently.

5. Budget and Operational Constraints

Teams must develop a redesign that:

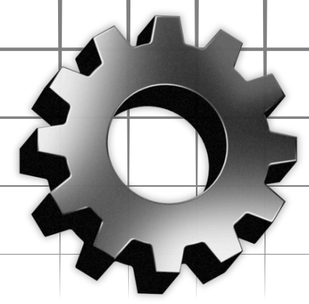
- Uses existing infrastructure and resources wherever practical.
- Minimizes additional operating costs.
- Maintains safe food handling and storage conditions.
- Accounts for limited transportation and refrigeration capacity.
- Provides backup procedures for further disruptions.
- Can be implemented rapidly during an emergency.

For this scenario, teams have a maximum additional emergency operating budget of $5,000 CAD to support their proposed response. This budget is intended for additional emergency expenses beyond the existing network's normal operating resources. Teams must identify and justify any proposed additional expenditures while prioritizing the use of available infrastructure, vehicles, personnel, and equipment.

6. Required Engineering Analysis

Teams must demonstrate how their proposed redesign improves emergency food distribution by addressing:

- Food allocation and prioritization.
- Transportation and distribution efficiency.
- Protection of perishable food.



- Resource utilization and operational costs.
- System reliability and backup procedures.
- Recovery of normal food distribution operations.

Teams must include at least one redesigned system or process diagram and provide a before-and-after comparison using relevant performance indicators or justified estimates.

Teams must account for the difference between the available perishable-food inventory and the available refrigerated-storage capacity.

Any proposed strategy for transporting, storing, redistributing, or temporarily managing perishable food must consider appropriate temperature control, handling conditions, available resources, and food-safety risks.

Teams must not assume that food remains safe solely because it has been relocated or placed in temporary storage.

Where necessary, teams should identify food that cannot be safely preserved or redistributed and explain the operational consequences of that decision.

Any additional assumptions must be clearly stated and supported by reasonable engineering justification.

Teams should consider food safety, transportation efficiency, resource availability, operational costs, emergency coordination, and implementation challenges when developing their redesign.

Teams must support their proposed solutions with appropriate system diagrams, performance comparisons, engineering justifications, and any calculations required by the official Case 2 scenario.

Note: Case 2 will be released during the competition briefing on Saturday, November 7, 2026. Teams will have eight (8) hours to work on the competition-day requirements and prepare their final deliverables.

Both cases must be submitted together on November 7, 2026, along with the final presentation, by the deadline communicated by the Competition Coordinators.